Grade 8 Practice

Standard 8.NS.1

Name: _____ Date: _____

Computation

1. Is 0.333... (repeating) a rational number? Write yes or no. **yes**

2. Is $\sqrt{2}$ a rational number? Write yes or no. **no**

3. Is 0.75 a rational number? Write yes or no. **yes**

4. Is π a rational number? Write yes or no. **no**

Word problems

5. Is $\sqrt{16}$ a rational number? Write yes or no.

Show your work and answer.

Answer: yes

6. Write 2/3 as a repeating decimal. _____

Show your work and answer.

Answer: 0.666...

Grade 8 Practice

Standard 8.NS.2

Name: _____ Date: _____

Computation

1. Between which two whole numbers is $\sqrt{10}$? _____ and _____

3 and 4

2. Between which two whole numbers is $\sqrt{20}$? _____ and _____

4 and 5

3. Approximate $\sqrt{2}$ to the nearest tenth. _____

1.4

4. Approximate $\sqrt{7}$ to the nearest hundredth. _____

2.65

Word problems

5. A square has an area of 50 square units. Between which two whole numbers is its side length?

Show your work and answer.

Answer: 7 and 8

6. Is $\sqrt{5}$ between 2.2 and 2.3? Write yes or no.

Show your work and answer.

Answer: yes

Grade 8 Practice

Standard 8.EE.1

Name: _____ Date: _____

Computation

1. Write $2^3 \times 2^2$ as a single power of 2. _____

2^5

2. Solve: 5^0 = _____

1

3. Write $(3^2)^3$ as a single power of 3. _____

3^6

4. Write $10^4 \div 10^2$ as a single power of 10. _____

10^2

Word problems

5. Write $x^4 \cdot x^3$ as a single power of x.

Show your work and answer.

Answer: x^7

6. Solve: 2^{-3} = _____

Show your work and answer.

Answer: 1/8

Grade 8 Practice

Standard 8.EE.2

Name: _____ Date: _____

Computation

1. Solve: $\sqrt{49}$ = _____ **7**

2. Solve: $\sqrt[3]{27}$ = _____ **3**

3. If $\sqrt{x} = 9$, what is x? _____ **81**

4. If $x^2 = 64$ and x is positive, what is x? _____ **8**

Word problems

5. A square has an area of 144 square feet. What is the side length?

Show your work and answer.

Answer: 12 feet

6. A cube has a volume of 125 cubic units. What is the side length?

Show your work and answer.

Answer: 5 units

Grade 8 Practice

Standard 8.EE.3

Name: _____ Date: _____

Computation

1. Write 3×10^4 as a standard number. _____

30,000

2. Write 5,000 in scientific notation. _____

5×10^3

3. Write 0.002 in scientific notation. _____

2×10^{-3}

4. Which is larger: 3×10^5 or 2×10^6? Write your answer.

2×10^6

Word problems

5. Earth's mass is about 6×10^{24} kg. The Moon's mass is about 7×10^{22} kg. Which is larger?

Show your work and answer.

Answer: Earth

6. Light travels 3×10^8 meters per second. How far does it travel in 10^2 seconds? Write your answer in scientific notation.

Show your work and answer.

Answer: 3×10^{10} meters

Grade 8 Practice

Standard 8.EE.4

Name: _____ Date: _____

Computation

1. Solve: $(2 \times 10^3)(3 \times 10^2) =$ _____

6 × 10⁵

2. Solve: $(8 \times 10^6) \div (2 \times 10^2) =$ _____

4 × 10⁴

3. Solve: $(4 \times 10^3) + (2 \times 10^3) =$ _____

6 × 10³

4. Solve: $(1.5 \times 10^4)(2 \times 10^1) =$ _____

3 × 10⁵

Word problems

5. Each cell has 3×10^6 parts. There are 4×10^2 cells. What is the total number of parts?

Show your work and answer.

Answer: 1.2 × 10⁹ parts

6. A trip is 9×10^7 miles at 3×10^2 miles per hour. How many hours does it take?

Show your work and answer.

Answer: 3 × 10⁵ hours

Grade 8 Practice

Standard 8.EE.5

Name: _____ Date: _____

Computation

1. What is the slope of the line through (0, 0) and (2, 6)? _____ **3**

2. What is the slope of the line through (1, 2) and (3, 6)? _____ **2**

3. What is the slope of the line through (0, 4) and (2, 0)? _____ **-2**

4. What is the slope of a horizontal line? _____ **0**

Word problems

5. A runner covers 4 miles in 0.5 hours. What is the speed in miles per hour?

Show your work and answer.

Answer: 8 mph

6. A graph passes through (1, 5) and (3, 15). What is its slope?

Show your work and answer.

Answer: 5

Grade 8 Practice

Standard 8.EE.6

Name: _____ Date: _____

Computation

1. What is the slope of y = 2x + 3? _____

2

2. What is the y-intercept of y = 2x + 3? _____

3

3. What is the slope of y = -x + 5? _____

-1

4. Write the equation of a line with slope 2 and y-intercept 4. _____

y = 2x + 4

Word problems

5. A phone plan costs $20 per month plus $5 per GB. Write an equation for the cost y after x GB.

Show your work and answer.

Answer: **y = 5x + 20**

6. A tank starts with 30 gallons and drains 2 gallons per hour. Write an equation for the gallons y after x hours.

Show your work and answer.

Answer: **y = -2x + 30**

Grade 8 Practice

Standard 8.EE.7

Name: _____ Date: _____

Computation

1. Solve: $3(x + 2) = 15$. $x =$ _____

3

2. Solve: $2x + 5 = 2x + 9$. Write the number of solutions.

no solution

3. Solve: $4x - 3 = 4x - 3$. Write the number of solutions.

infinitely many

4. Solve: $5x - 2 = 13$. $x =$ _____

3

Word problems

5. Plan A costs $20 + 5x$ and plan B costs $30 + 3x$. For what x are the costs equal?

Show your work and answer.

Answer: **$x = 5$**

6. Solve: $2(x - 4) = 3x$. $x =$ _____

Show your work and answer.

Answer: **$x = -8$**

Grade 8 Practice

Standard 8.EE.8

Name: _____ Date: _____

Computation

1. Solve the system: y = x + 1 and y = 3. Write (x, y).

(2, 3)

2. Solve the system: y = 2x and y = x + 4. Write (x, y).

(4, 8)

3. Solve the system: x + y = 10 and x - y = 2. Write (x, y).

(6, 4)

4. Solve the system: y = -x and y = x. Write (x, y).

(0, 0)

Word problems

5. Plan A costs 10 + 2x and plan B costs 4x. For what x are the costs equal?

Show your work and answer.

Answer: x = 5

6. Solve the system: x + y = 12 and y = x + 2. What is x?

Show your work and answer.

Answer: x = 5

Grade 8 Practice

Standard 8.F.1

Name: _____ Date: _____

Computation

1. Table: x = 1, 2, 3 gives y = 4, 4, 5. Is this a function? Write yes or no. **yes**

2. Points (1, 2) and (1, 3) are in a relation. Is it a function? Write yes or no. **no**

3. Is $y = x^2$ a function? Write yes or no. **yes**

4. Is the vertical line x = 4 a function? Write yes or no. **no**

Word problems

5. Each student has exactly one locker number. Is this a function? Write yes or no.

Show your work and answer.

Answer: yes

6. Each student's name is matched with a shoe size. Is this a function? Write yes or no.

Show your work and answer.

Answer: yes

Grade 8 Practice

Standard 8.F.2

Name: _____ Date: _____

Computation

1. Function A is y = 2x + 1. Function B has y = 3 when x = 0. Which has the greater y-intercept? Write A or B.

B

2. Function A has slope 2. Function B has slope 3. Which is steeper? Write A or B.

B

3. A line passes through (0, 2) and (2, 6). What is its slope? _____

2

4. Function y = 3x + 2 has what y-intercept? _____

2

Word problems

5. Plan A costs $20 plus $3 per hour. Plan B costs $5 plus $6 per hour. Which plan has the lower starting cost? Write A or B.

Show your work and answer.

Answer: B

6. Plan A earns $6 per hour and Plan B earns $5 per hour. Which plan has the greater rate? Write A or B.

Show your work and answer.

Answer: A

Grade 8 Practice

Standard 8.F.3

Name: _____ Date: _____

Computation

1. Is $y = x^2$ a linear function? Write yes or no.

no

2. Is $y = 3x + 1$ a linear function? Write yes or no.

yes

3. Is the area of a square, $A = s^2$, a linear function of side length? Write yes or no.

no

4. Is $y = 2^x$ a linear function? Write yes or no.

no

Word problems

5. The volume of a cube, $V = s^3$, is a function of side length. Is it linear? Write yes or no.

Show your work and answer.

Answer: **no**

6. The circumference $C = 2\pi r$ is a function of radius. Is it linear? Write yes or no.

Show your work and answer.

Answer: **yes**

Grade 8 Practice

Standard 8.F.4

Name: _____ Date: _____

Computation

1. A rate is 4 per hour and starts at 0. Write the equation.

y = 4x

2. Start at 10 and add 3 per minute. Write the equation using y and x.

y = 3x + 10

3. A table has points (1, 5) and (2, 7). Write the linear equation.

y = 2x + 3

4. Write the equation for the area of a square in terms of side length s.

$y = s^2$

Word problems

5. A phone plan costs $25 plus $10 per month. Write an equation for the cost y after m months.

Show your work and answer.

Answer: **y = 10x + 25**

6. A car uses 2 gallons for every 60 miles. Write an equation for gallons y used after x miles.

Show your work and answer.

Answer: **y = x/30**

Grade 8 Practice

Standard 8.F.5

Name: _____ Date: _____

Computation

1. A graph rises steadily from left to right. Describe it as increasing or decreasing.

increasing

2. A graph stays flat. Describe it as constant or changing.

constant

3. A graph falls from left to right. Describe it as increasing or decreasing.

decreasing

4. A graph is a straight line through the origin going up. Is it increasing or decreasing?

increasing

Word problems

5. Water fills a tub, then drains. Describe the graph in words.

Show your work and answer.

Answer: **Increases, then decreases**

6. A bike rides at a constant speed. What does its distance-time graph look like?

Show your work and answer.

Answer: **straight line**

Grade 8 Practice

Standard 8.G.1

Name: _____ Date: _____

Computation

1. Reflect (2, 3) over the x-axis. _____ **(2, -3)**

2. Translate (1, 1) 3 units right. _____ **(4, 1)**

3. Rotate (1, 0) 90° counterclockwise about the origin. _____ **(0, 1)**

4. Dilate (2, 4) by a factor of 2 from the origin. _____ **(4, 8)**

Word problems

5. A triangle vertex at (3, 1) is reflected over the y-axis. What is the new point?

Show your work and answer.

Answer: (-3, 1)

6. Point (-2, 5) is translated 4 units down. What is the new point?

Show your work and answer.

Answer: (-2, 1)

Grade 8 Practice

Standard 8.G.2

Name: _____ Date: _____

Computation

1. Two figures have the same size and shape. Are they congruent? Write yes or no.

yes

2. Is a figure congruent to its dilation? Write yes or no.

no

3. Is a rotated figure congruent to the original? Write yes or no.

yes

4. Is a stretched figure congruent to the original? Write yes or no.

no

Word problems

5. Two triangles have sides 3, 4, and 5. A second triangle has the same side lengths. Are they congruent?

Show your work and answer.

Answer: yes

6. A rectangle is dilated by a factor of 2. Is the new rectangle congruent to the original?

Show your work and answer.

Answer: no

Grade 8 Practice

Standard 8.G.3

Name: _____ Date: _____

Computation

1. Reflect (-4, 2) over the y-axis. _____ **(4, 2)**

2. Translate (0, 0) by (5, -2). _____ **(5, -2)**

3. Rotate (2, 3) 180° about the origin. _____ **(-2, -3)**

4. Dilate (3, -1) by a factor of 3 from the origin. _____ **(9, -3)**

Word problems

5. Point (1, 6) is reflected over the x-axis. What is the new point?

Show your work and answer.

Answer: (1, -6)

6. Point (4, 0) is rotated 90° counterclockwise about the origin. What is the new point?

Show your work and answer.

Answer: (0, 4)

Grade 8 Practice

Standard 8.G.4

Name: _____ Date: _____

Computation

1. Are all squares similar? Write yes or no.

yes

2. Are rectangles 2 by 3 and 4 by 6 similar? Write yes or no.

yes

3. Are rectangles 2 by 3 and 2 by 4 similar? Write yes or no.

no

4. Are two triangles with equal angles similar? Write yes or no.

yes

Word problems

5. A 3 by 4 rectangle is dilated by a factor of 2. Is the new rectangle similar to the original? Write yes or no.

Show your work and answer.

Answer: **yes**

6. A segment is 5 units long and its image is 15 units long. What is the scale factor?

Show your work and answer.

Answer: **3**

Grade 8 Practice

Standard 8.G.5

Name: _____ Date: _____

Computation

1. Two triangles have angles 40°, 60°, 80° and 40°, 60°, 80°. Are they similar? Write yes or no.

yes

2. A triangle has remote interior angles 60° and 70°. What is the exterior angle? _____

130°

3. What is the sum of the angles in a triangle? _____

180°

4. Two parallel lines are cut by a transversal. Are alternate interior angles equal? Write equal or not equal.

equal

Word problems

5. A triangle has angles 50° and 70°. What is the third angle?

Show your work and answer.

Answer: 60°

6. On parallel lines, a 65° angle corresponds to what other angle measure?

Show your work and answer.

Answer: 65°

Grade 8 Practice

Standard 8.G.6

Name: _____ Date: _____

Computation

1. A right triangle has legs 3 and 4. What is the hypotenuse? _____

5

2. A right triangle has legs 6 and 8. What is the hypotenuse? _____

10

3. A right triangle has hypotenuse 13 and one leg 5. What is the other leg? _____

12

4. A right triangle has legs 5 and 12. What is the hypotenuse? _____

13

Word problems

5. A 10-foot ladder reaches 6 feet up a wall. How far is the base of the ladder from the wall?

Show your work and answer.

Answer: 8 feet

6. A square has sides of 5. What is the length of its diagonal? Write your answer as a square root.

Show your work and answer.

Answer: $\sqrt{50}$

Grade 8 Practice

Standard 8.G.7

Name: _____ Date: _____

Computation

1. Right triangle legs 9 and 12. What is the hypotenuse? _____ **15**

2. Right triangle hypotenuse 17 and one leg 8. What is the other leg? _____ **15**

3. Right triangle with legs 1 and 1. What is the hypotenuse? Write your answer as a square root. _____ **$\sqrt{2}$**

4. Right triangle hypotenuse 10 and one leg 6. What is the other leg? _____ **8**

Word problems

5. A box is 3 by 4 by 12 units. What is the length of its longest diagonal?

Show your work and answer.

Answer: 13 units

6. A floor is 6 feet by 8 feet. What is the diagonal of the floor?

Show your work and answer.

Answer: 10 feet

Grade 8 Practice

Standard 8.G.8

Name: _____ Date: _____

Computation

1. Find the distance from (0, 0) to (3, 4). _____ **5**

2. Find the distance from (1, 1) to (4, 5). _____ **5**

3. Find the distance from (-2, 0) to (2, 0). _____ **4**

4. Find the distance from (0, 2) to (3, 6). _____ **5**

Word problems

5. A park is at (0, 0) and a school is at (6, 8). How far apart are they?

Show your work and answer.

Answer: 10 units

6. A ball moves from (1, 2) to (4, 6). How far does it travel in a straight line?

Show your work and answer.

Answer: 5 units

Grade 8 Practice

Standard 8.G.9

Name: _____ Date: _____

Computation

1. A cylinder has radius 2 and height 5. Using π ≈ 3.14, find its volume. _____

62.8

2. A cone has radius 3 and height 4. Using π ≈ 3.14, find its volume. _____

37.68

3. A sphere has radius 3. Using π ≈ 3.14, find its volume. _____

113.04

4. A cylinder has radius 1 and height 10. Using π ≈ 3.14, find its volume. _____

31.4

Word problems

5. A can has a radius of 4 cm and a height of 10 cm. Using π ≈ 3.14, what is its volume?

Show your work and answer.

Answer: 502.4 cubic cm

6. A cone has a radius of 2 cm and a height of 6 cm. Using π ≈ 3.14, what is its volume?

Show your work and answer.

Answer: 25.12 cubic cm